Predictive Resource Optimization in Pharmacy Operations

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Executive Summary

This project develops an end to end analytics pipeline to forecast prescription demand, evaluate staffing adequacy, and analyze revenue performance within a simulated pharmacy operation. Using Python, SQL, R, and Power BI, the system integrates demand forecasting, operational metrics, and financial analytics into a unified decision support framework.

Key outcomes include improved visibility into peak workload periods, identification of staffing imbalances, and enhanced understanding of revenue distribution across insurance types and drug classes. The resulting dashboard provides pharmacy leadership with actionable insights for workforce planning and inventory strategy.

Business Problem

Pharmacy operations must balance fluctuating prescription demand, limited staffing resources, and complex payer mixes. Understaffing risks service delays and patient dissatisfaction, while overstaffing increases labor costs. Simultaneously, inventory and revenue performance must be monitored to maintain financial sustainability.

This project addresses the need for an integrated analytics solution that:

- Forecasts prescription demand at the drug class level
- Evaluates hourly staffing adequacy
- Quantifies operational gaps
- Analyzes revenue performance by payer mix

Data Sources

The project combines real reference data with synthetic operational data.

Real Data

- OpenFDA NDC Directory for drug product reference information
  https://open.fda.gov/data/ndc/

Synthetic Data

Synthetic datasets were generated to simulate pharmacy operations, including:

- Patient population and payer mix
- Prescription transaction history
- Inventory snapshots
- Pharmacist staffing schedules
- Sales and revenue data

Synthetic generation enables controlled experimentation while preserving realistic operational patterns and protecting patient privacy.

Data Engineering Pipeline

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Notebook 1: Core Data Generation

This stage constructs the foundational operational datasets. These datasets establish the transactional and reference backbone of the pipeline.

Outputs:

- patients.csv
- products.csv
- prescriptions.csv
- inventory.csv
- sales.csv
- staffschedule.csv
- ref_drugs.csv
- prescriptions_by_class.csv

Notebook 2: Forecasting and Feature Engineering

This stage develops demand signals and predictive features at the drug class level. The forecasting framework captures temporal demand patterns and supports forward looking planning.

Outputs:

- predictions_daily_class.csv
- forecast_demand_by_class_day.csv
- daily_demand_by_class.csv
- inventory_suggestions_by_class.csv
- daily_class_features.csv
- model_metrics_by_class.csv
- forecast_6m_by_class.csv

Notebook 3: Staffing Optimization

This component evaluates workforce alignment with workload demand.

Key steps include:

- Construction of hourly prescription workload
- Estimation of required staffing levels
- Reconstruction of scheduled staffing from weekly schedules
- Computation of overstaffed and understaffed intervals

Primary output:

- projected_staffing_by_hour.csv

Notebook 4: Business Intelligence Layer

This stage produces a clean semantic layer for Power BI consumption. These tables support the analytical dashboard.

Outputs:

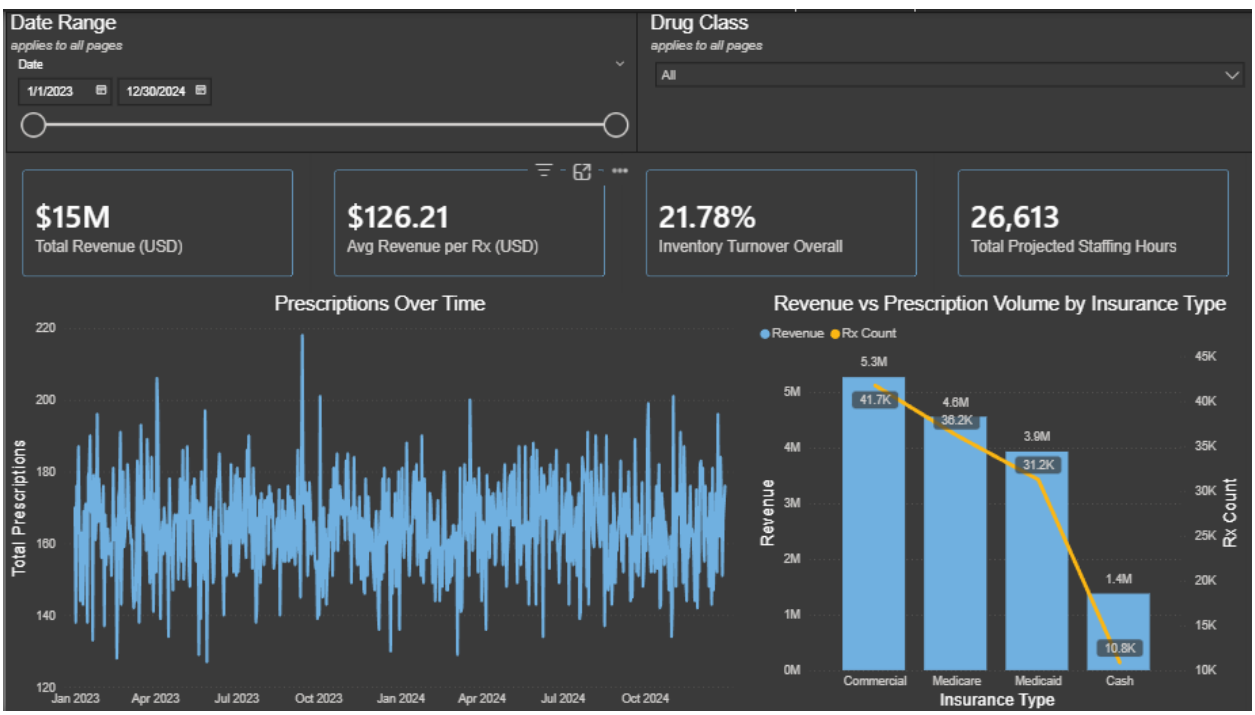
- date_table.csv
- prescription_volume_by_day.csv
- prescription_volume_by_class_day.csv
- forecast_demand_by_class_day.csv
- inventory_turnover.csv
- kpi_summary.csv
- projected_staffing_by_hour.csv

Dashboard Design

The Power BI dashboard is organized into five analytical views:

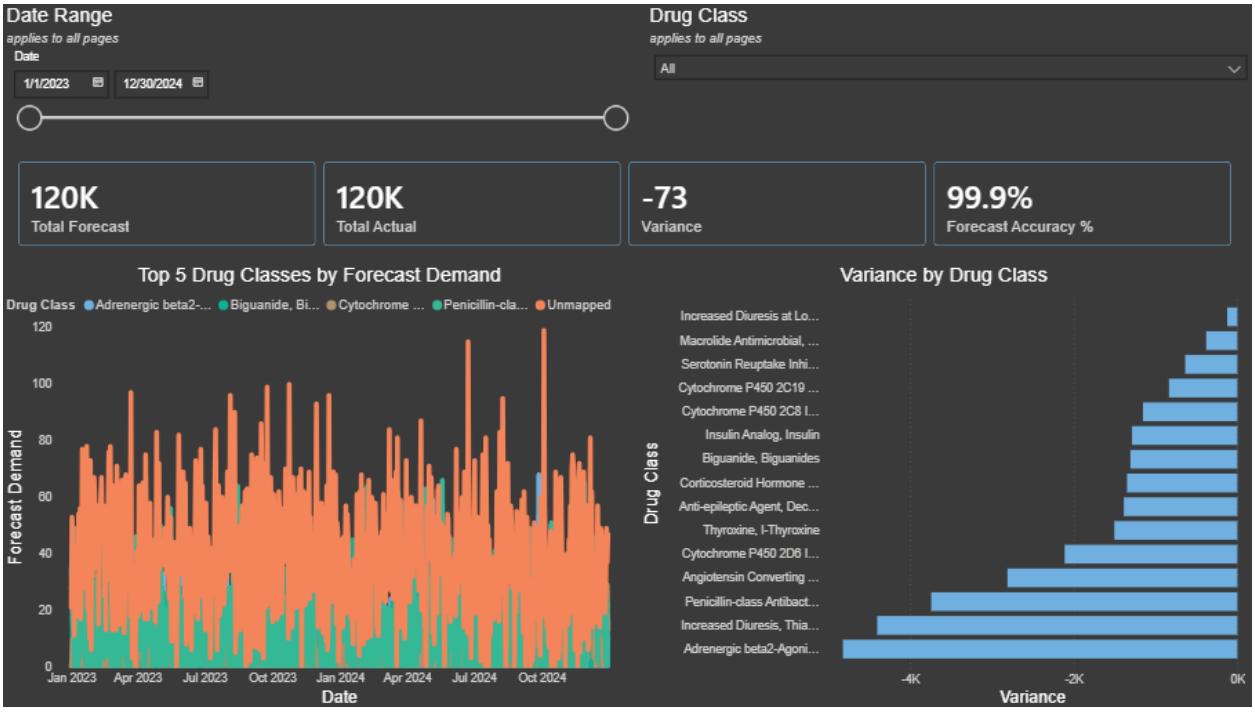
1) Executive Overview

Provides high level KPIs including total prescriptions, forecast accuracy, inventory turnover, and revenue metrics.



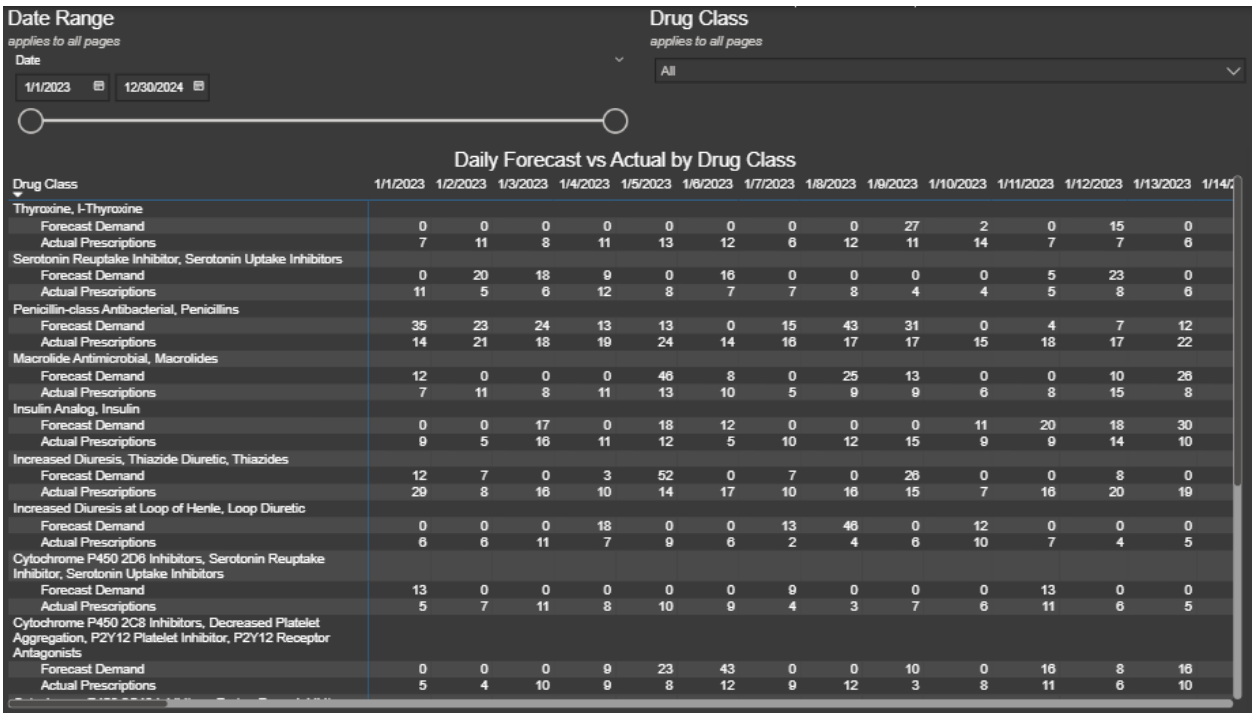
2) Forecast Analysis

Visualizes demand trends and forecast variance across drug classes.



3) Detailed Class Breakdown

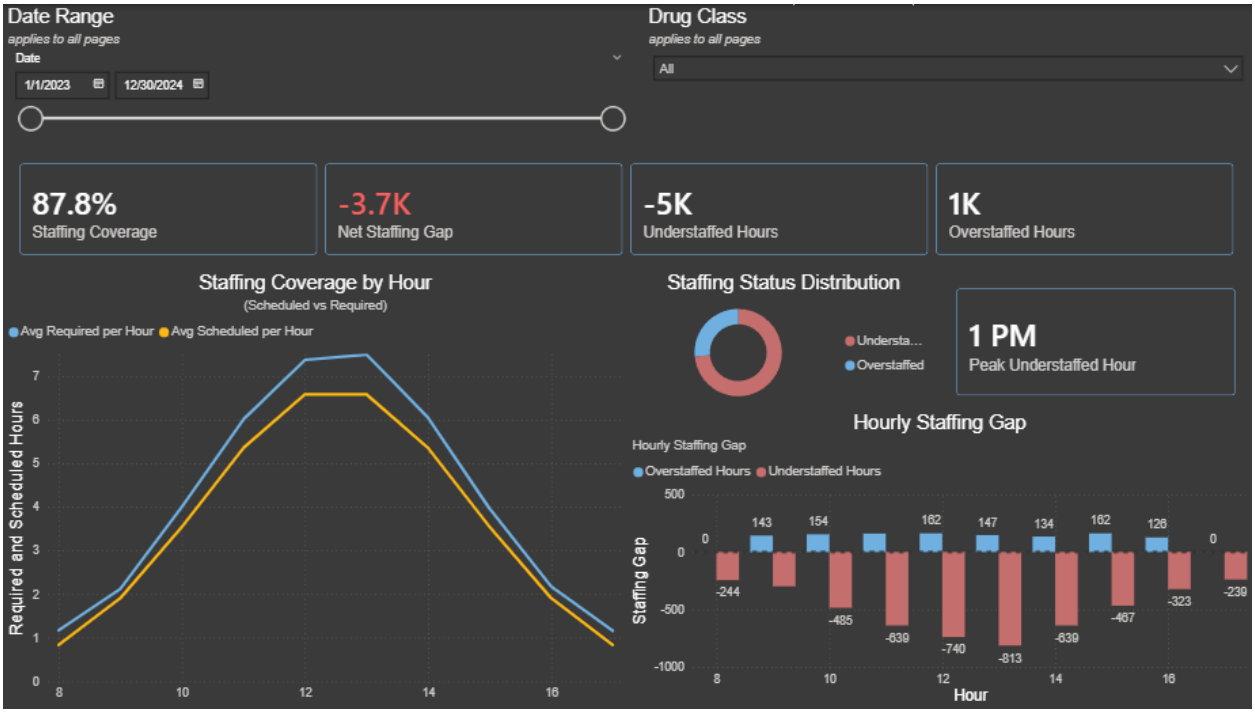
Delivers class level visibility into prescription volume, forecast accuracy, and demand variability across therapeutic categories.



Drug Class	1/1/2023	1/2/2023	1/3/2023	1/4/2023	1/5/2023	1/6/2023	1/7/2023	1/8/2023	1/9/2023	1/10/2023	1/11/2023	1/12/2023	1/13/2023	1/14/2
Thyroxine, l-Thyroxine														
Forecast Demand	0	0	0	0	0	0	0	0	27	2	0	15	0	
Actual Prescriptions	7	11	8	11	13	12	6	12	11	14	7	7	6	
Serotonin Reuptake Inhibitor, Serotonin Uptake Inhibitors														
Forecast Demand	0	20	18	9	0	16	0	0	0	0	5	23	0	
Actual Prescriptions	11	5	6	12	8	7	7	8	4	4	5	8	6	
Penicillin-class Antibacterial, Penicillins														
Forecast Demand	35	23	24	13	13	0	15	43	31	0	4	7	12	
Actual Prescriptions	14	21	18	19	24	14	16	17	17	15	18	17	22	
Macrolide Antimicrobial, Macrolides														
Forecast Demand	12	0	0	0	46	8	0	25	13	0	0	10	26	
Actual Prescriptions	7	11	8	11	13	10	5	9	9	6	8	15	8	
Insulin Analog, Insulin														
Forecast Demand	0	0	17	0	18	12	0	0	0	11	20	18	30	
Actual Prescriptions	9	5	16	11	12	5	10	12	15	9	9	14	10	
Increased Diuresis, Thiazide Diuretic, Thiazides														
Forecast Demand	12	7	0	3	52	0	7	0	26	0	0	8	0	
Actual Prescriptions	29	8	16	10	14	17	10	16	15	7	16	20	19	
Increased Diuresis at Loop of Henle, Loop Diuretic														
Forecast Demand	0	0	0	18	0	0	13	46	0	12	0	0	0	
Actual Prescriptions	6	6	11	7	9	6	2	4	6	10	7	4	5	
Cytochrome P450 2D6 Inhibitors, Serotonin Reuptake Inhibitor, Serotonin Uptake Inhibitors														
Forecast Demand	13	0	0	0	0	0	9	0	0	0	13	0	0	
Actual Prescriptions	5	7	11	8	10	9	4	3	7	6	11	6	5	
Cytochrome P450 2C8 Inhibitors, Decreased Platelet Aggregation, P2Y12 Platelet Inhibitor, P2Y12 Receptor Antagonists														
Forecast Demand	0	0	0	9	23	43	0	0	10	0	16	8	16	
Actual Prescriptions	5	4	10	9	8	12	9	12	3	8	11	6	10	

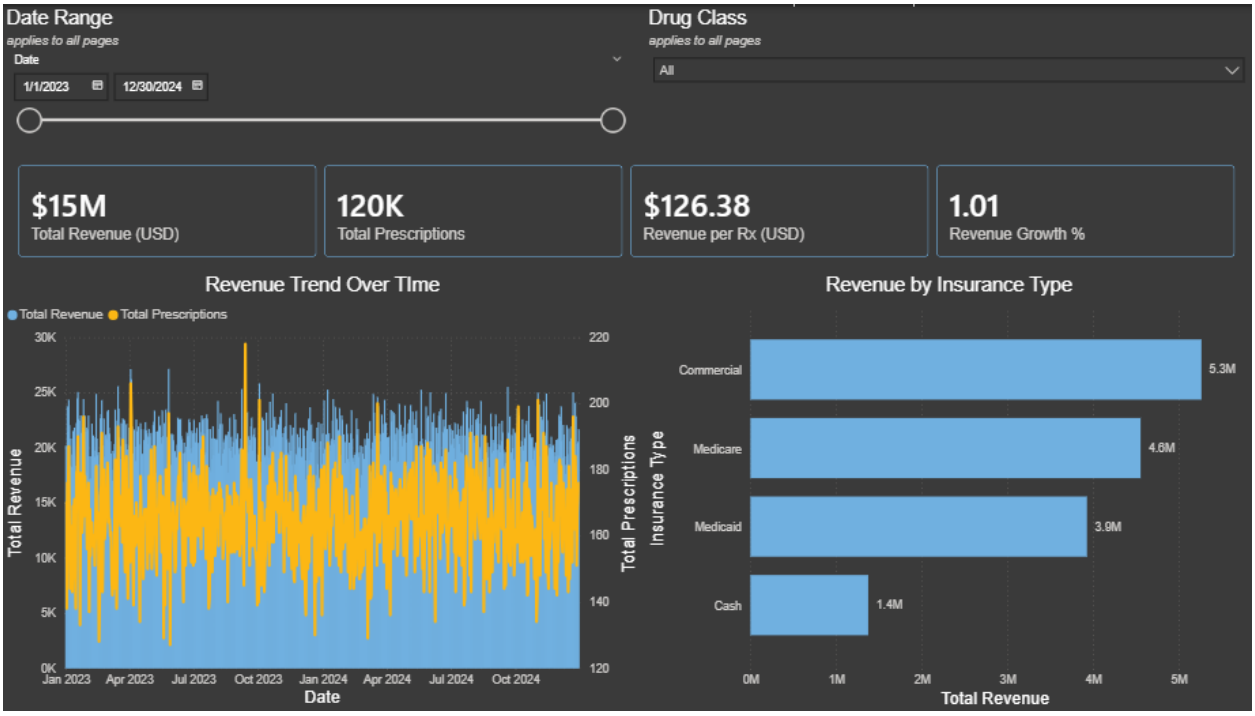
4) Operational Efficiency and Staffing

Evaluates hourly staffing coverage and identifies periods of operational risk.



5) Revenue Performance

Analyzes revenue distribution by insurance type and prescription volume relationships.



Key Findings

- Prescription demand exhibits meaningful variation across drug classes.
- Several peak hours show sustained understaffing risk.
- Revenue is concentrated among a subset of payer types.
- Inventory turnover varies significantly by therapeutic category.

These findings demonstrate the value of integrated operational analytics in pharmacy environments.

Limitations

- Staffing schedules are synthetically generated.
- Demand model uses simplified smoothing assumptions.
- Service time per prescription is treated as constant.
- External seasonality drivers were not incorporated.

Future iterations should incorporate real operational telemetry.

Future Work

Potential enhancements include:

- Queueing based service time modeling
- Real time staffing feeds
- Advanced time series forecasting methods
- Automated data pipeline orchestration

Conclusion

This project demonstrates how integrated data engineering, forecasting, and business intelligence can provide actionable operational insight within pharmacy workflows. The resulting framework supports more informed staffing, inventory, and financial decision making.